

Genetic Information: Cell to Gene

LAUNCH • Science • 40 minutes

I can describe where genetic information is stored and how chromosomes, DNA, genes and the genome are related.

Prepare before pupils arrive

Setup: Prepare large nested cards and a DNA ribbon that can be placed inside the chromosome card.

Safety: Avoid tiny model components where they create choking, sensory or fine-motor barriers.

Equipment: Nested hierarchy cards, chromosome/DNA ribbon model, labels, arrow strip and pencil.

Safety: Desk-based. Use large pieces rather than small loose beads where choking or fine-motor access is a concern.

Fallback: Direct an adult to place each large card and dictate the contains/section-of links.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe DNA, genes and chromosomes.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 4 min	State where chromosomes are found in a eukaryotic cell. Recall why daughter cells receive identical chromosome sets in mitosis.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do • 5 min	chromosome — packaged DNA gene — section of DNA genome — all an organism's DNA Start with a eukaryotic body cell and locate its nucleus. Inside the nucleus, identify chromosomes; each chromosome contains one long DNA molecule associated with proteins.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	A gene's base sequence provides information for making a particular protein; the lesson does not require Higher-tier protein-synthesis stages. Different genes occupy different sections of chromosome DNA, while the genome includes all chromosomes and other DNA in the organism.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each description at the correct level, then read the complete hierarchy aloud or by pointing. CELL NUCLEUS CHROMOSOME DNA GENE Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Build and annotate the genetic-information hierarchy, then explain one model strength and one limitation. Record: An ordered hierarchy, accurate definitions and one honest statement about scale or shape.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Describe how a chromosome, DNA and a gene are related. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: A gene is a section of DNA located on a chromosome.

Yes. A chromosome contains DNA, and a gene is a section of that DNA.

Misconception repair

Repaired. A chromosome contains a long DNA molecule, and each gene is a section of that DNA.

Guided checkpoint

Yes. A chromosome contains DNA, and a gene is a section of that DNA. Ask which item is the whole molecule and which is a smaller section.

Location

CELL / NUCLEUS LEVEL. Nucleus placed as the chromosome-containing compartment. Look for the compartment inside the cell.

Long molecule

CHROMOSOME / DNA LEVEL. DNA placed at the chromosome-molecule level. Which molecule forms the genetic material of a chromosome?

Coding section

GENE / GENOME IDEA. Gene placed as a section of DNA. Look for the smaller section, not the whole chromosome.

Model limitation

The cards are not to scale, omit most molecular detail and can wrongly imply chromosomes are always X-shaped if the limitation is not stated.

Independent evidence

An ordered hierarchy, accurate definitions and one honest statement about scale or shape.

Supported: Order five icon-backed cards and complete: a gene is a section of ___. Two cards are pre-positioned; use contains and part of arrows.

Standard: Build the full hierarchy and write two linked relationship sentences. Use: The nucleus contains ___. A gene is ___.

Stretch: Explain the hierarchy, include genome and critique the permanent-X and scale misconceptions. Keep to Foundation content; depth comes from relational precision and model evaluation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.

What the learner actually showed	Next teaching action
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_L_W12L1_DNA_Hierarchy_Introduce.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.